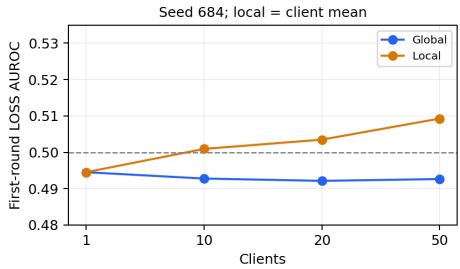
FedAvg and first-round membership inference

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- 9,000 synthetic encounters; Assignment 1 generation pipeline reused.
- 50/50 split: 3,600 SGD + 900 validation / 4,500 test.
- 28 FedAvg settings × 3 seeds; 300-round cap.
- L2 + validation scheduler; best validation checkpoint restored.
- **Central synthetic validation is a simulation assumption.**
- **72 loss-stable; 9 early stops; 3 capped.**
- Baseline AUROC 0.657–0.707; validation-threshold BA 0.606–0.638.
- $K = 1$ test CE: 1.275 → 0.323; moderate predictive utility.



Plot: seed 684 only. Listed metrics are three-seed means. All 84 runs and 1,785 attacks audited. First-round LOSS remains approximately chance; this is not a privacy guarantee.

Local nonmembers: all records outside that client's optimization partition. Oracle BA chooses its threshold retrospectively.
Secure aggregation hides individual updates; differential privacy limits membership leakage.